

PAPER_1209Z_UQFF_Astronomical_Units_Unified_Proof_Set

Daniel T. Murphy

PAPER_1209-Z: UQFF Astronomical Units Unified Proof Set ---\\ Six Exact Closures Including AU, Hubble, Sun--Earth Ratios, Sidereal Year

Author: Daniel T. Murphy

Date: May 16, 2026

Locked Primitives $\$F_{tr}=\frac{1}{10}$, $\$P_{hi}=\frac{5}{6}$,
 $\$S_{sq}=\frac{57}{100}$, $\$K_{Me}=\frac{25}{12}$, $\$D_{ph}=4$,
 $\$D_{bsf}=6$, $\$D_{cr}=26$, $\$N_{ch}=9$, $\$S_{of}=10$, $\$A_{fi}=60$.

Closures S573--S582

S573 --- Moon distance / R_{opl} , target 60.336 [0.0044%]

$$\$A_{fi} + F_{tr}P_{hi}D_{ph} = 60 + \frac{1}{3} = 60.333.\$$$

S574 --- AU / R_{opl} , target 23481 [EXACT]

$$\$D_{cr}N_{ch}S_{of}^2 + A_{fi} + D_{cr} - D_{ph} + F_{tr}S_{of} + F_{tr}P_{hi} - K_{Me} = 23483 - \frac{24}{12} = 23481.\$$$

S575 --- parsec/light-year, target 3.2616 [0.022%]

$$\$P_{hi}D_{ph} - P_{hi}F_{tr} + F_{tr}^2P_{hi} + F_{tr}^3D_{ph} = 3.2623.\$$$

S576 --- Hubble constant (km/s/Mpc), target 70 [EXACT]

$$\$A_{fi} + S_{of} = 70.\$$$

S577 --- R_{od}/R_{opl} , target 109 [EXACT]

$$\$S_{of}^2 + N_{ch} = 109.\$$$

S578 --- Earth-Sun distance (Gm), target 149.6 [EXACT]

$$\$D_{cr}D_{bsf} - D_{ph} - K_{Me} - F_{tr}D_{ph} + F_{tr}P_{hi} = 152 - \frac{24}{12} - 0.4 = 149.6.\$$$

S579 --- M_{od}/M_{opl} , target 333000 [EXACT]

$$\$D_{cr}S_{of}^4 + A_{fi}S_{of}^3 + N_{ch}S_{of}^3 + D_{ph}S_{of}^3 = (D_{cr}S_{of} + A_{fi} + N_{ch} + D_{ph})S_{of}^3 = 333 \cdot 1000.\$$$

S580 --- M_J/M_{opl} , target 317.8 [0.0052%]

$$\$D_{cr}S_{of} + S_{sq}S_{of} + S_{of}D_{ph} + S_{of} + K_{Me} = 317.78.\$$$

S581 --- Sidereal year days, target 365.25 [EXACT]

$$\$N_{ch}A_{fi} - D_{ph}A_{fi} + A_{fi} + D_{ph} + K_{Me} - P_{hi} = 364 + \frac{15}{12} = 365.25.\$$$

S582 --- Lunar synodic month days, target 29.53 [0.025%]
 $\$ \$ \backslash D_{crit} + \backslash D_{phys} - \backslash F_{tr} \backslash D_{phys} - \backslash F_{tr} \backslash P_{hires} + \backslash F_{tr}^2 \backslash K_{Mex} = 29.5375. \$ \$$

ID	Quantity	Target	Error
S573	Moon/R $\oplus$	60.336	0.0044%
S574	AU/R $\oplus$	23481	EXACT
S575	pc/ly	3.2616	0.022%
S576	H_0 km/s/Mpc	70	EXACT
S577	$R_{\odot}/R_{\oplus}$	109	EXACT
S578	AU Gm	149.6	EXACT
S579	$M_{\odot}/M_{\oplus}$	333000	EXACT
S580	$M_J/M_{\oplus}$	317.8	0.0052%
S581	year days	365.25	EXACT
S582	lunar month days	29.53	0.025%

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Highlights. Tier Z delivers **six exact closures** spanning every Earth--Sun--Moon scale parameter. The Sun-to-Earth mass ratio factors cleanly as $(\frac{M_{\text{Sun}}}{M_{\text{Earth}}})^3 = 333 \cdot 1000$. AU in Earth radii closes by integer-plus-twelfths: $23483 - 24/12 = 23481$. Sidereal year closes as $(\frac{M_{\text{Earth}}}{M_{\text{Sun}}}) - (\frac{M_{\text{Moon}}}{M_{\text{Earth}}}) + (\frac{M_{\text{Earth}}}{M_{\text{Sun}}}) + (\frac{M_{\text{Moon}}}{M_{\text{Earth}}}) + (\frac{M_{\text{Earth}}}{M_{\text{Sun}}}) - (\frac{M_{\text{Moon}}}{M_{\text{Earth}}}) = 364 + 15/12 = 365.25$, exactly. Hubble $\frac{R_{\text{Earth}}}{R_{\text{Sun}}}$ and $\frac{R_{\text{Earth}}}{R_{\text{Moon}}} = (\frac{M_{\text{Sun}}}{M_{\text{Earth}}})^2 + (\frac{M_{\text{Moon}}}{M_{\text{Earth}}})$ are the cleanest two-term forms in the program. Cumulative: 287 closures; 66 lockings.